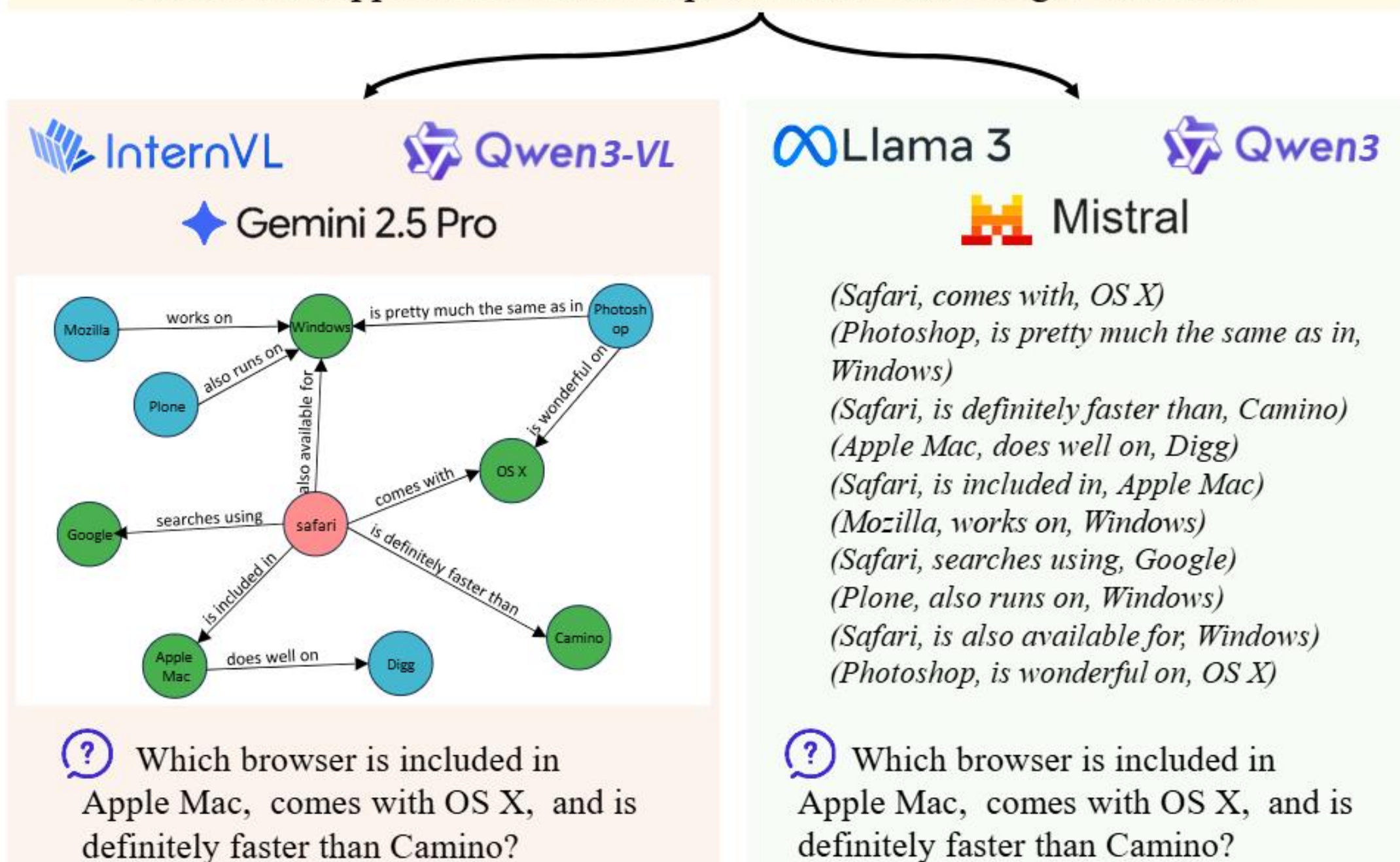


Motivation & Contributions

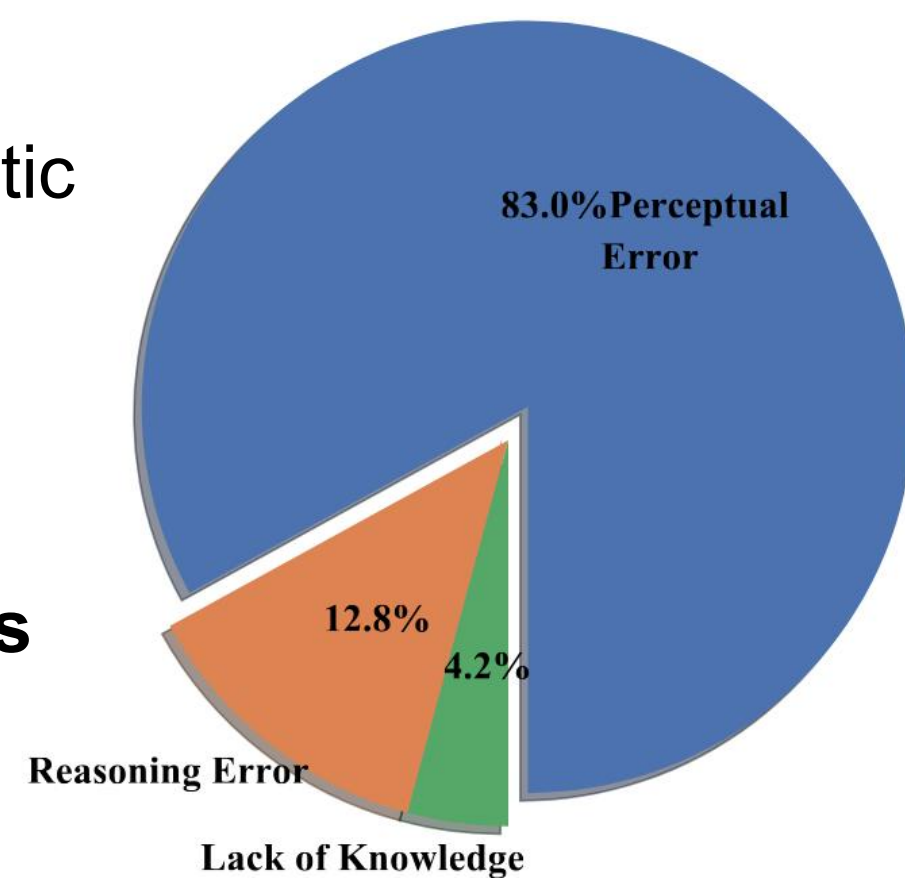
- KGs encode entities and relations explicitly, but textual linearization weakens **higher-order relational cues**.
- LMMs can naturally perceive visual layouts and spatial relations, offering a promising way to **understand KGs as visualized graphs**.

This is a knowledge graph showing that the Safari web browser is available for OS X and Windows, is faster than Camino, and includes features like Private Browsing while using Google for searches. It connects Safari to other entities like Apple Mac, Photoshop, Mozilla, and Google Checkout.



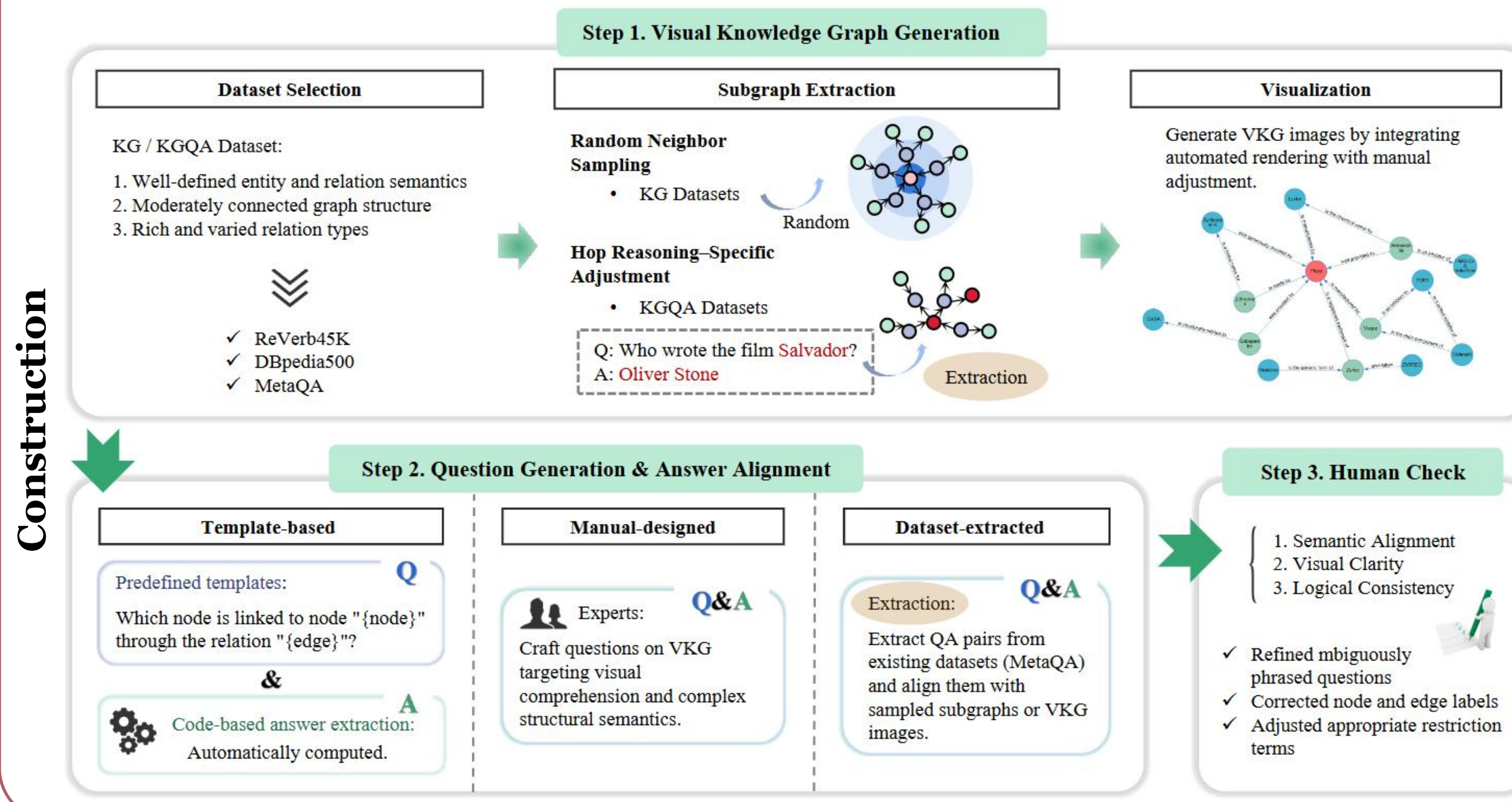
Contributions

- We introduce VKG-QA, a benchmark explicitly **designed to evaluate LMMs' capability to understand and reason over visualized KGs**, bridging the gap between graph reasoning and multimodal understanding.
- VKG-QA covers three categories and fourteen subtasks**, providing a systematic and fine-grained evaluation of structured multimodal reasoning, supported by a scalable semi-automatic benchmark construction pipeline.
- We conduct comprehensive **experiments on 19 state-of-the-art LMMs**, analyze their reasoning behaviors, and uncover critical limitations in structural perception and relational reasoning, offering insights for future graph-aware multimodal model design.



VKG-QA

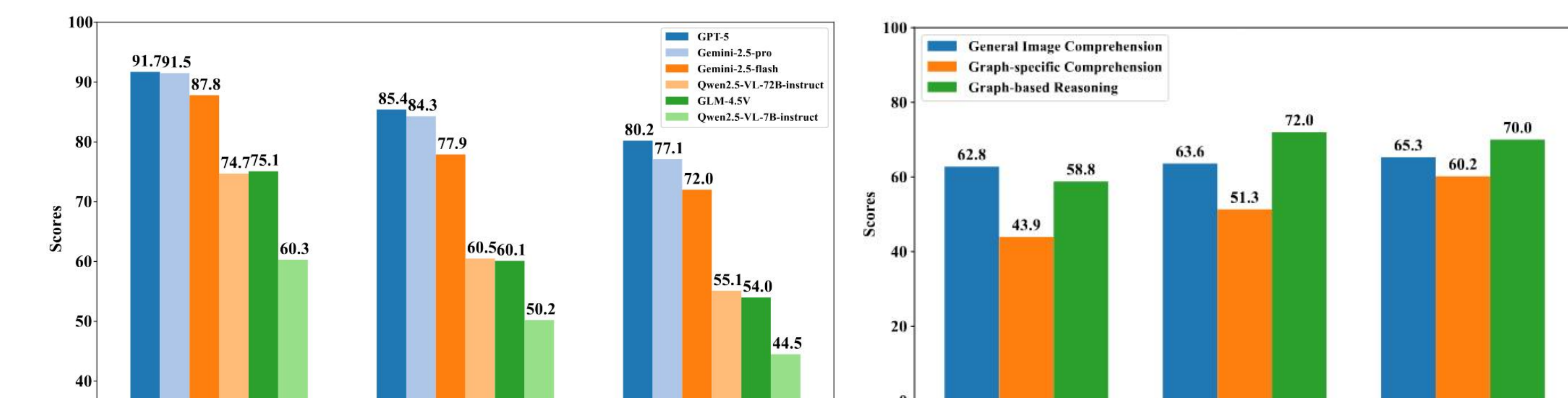
General image Comprehension	Graph-specific Comprehension
Q: How many different colors are there in the nodes of the given image? A: 3 <i>Color identification</i>	Q: Which node is linked to node 'Stewart' through the relation 'says of'? A: Leffler <i>Graph understand</i>
Q: In the given image, find the nodes whose names start with 'D'. A: Diego Maradona <i>Existence judgment</i>	Q: What is the out-degree of the node 'Kennedy'? A: 1 <i>Degree analysis</i>
Q: Count the total number of nodes in the given image. A: 28 <i>Basic counting</i>	Q: Which two nodes are connected by the arrow with label 'travelled to'? A: Lee Harvey Oswald, Moscow <i>relation direction identification</i>
Q: Return the name of the node that is at the upper left of point 'Almagro' and is green. A: China <i>Spatial position recognition</i>	Q: Can you start at node 'Rehnquist' and return to it without repeating any edges or nodes? A: no <i>Cycle detection</i>
Q: Extract all text appearing on the nodes in the given image. A: Russia, Sealand, The Pirate Bay <i>Text extraction</i>	Q: Does the set of nodes: '{Taliban}' constitute a strongly connected subgraph? A: yes <i>Connectivity assessment</i>
Graph-based Reasoning	
Q: When was the movie The Art of War released? A: 2000 <i>Single-hop reasoning</i>	Q: Which languages are spoken in films written by writers My Winnipeg? A: Icelandic <i>Multi-hop reasoning</i>
Q: Which album has the most producers credited to it? A: Strangers <i>Superlative reasoning</i>	Q: Which forward is from Honduras? A: José Güity <i>Conditional constraint reasoning</i>



Results

	Color	Exist	Count	Locat	Text	G-un.	Degree	Dire.	Cycle	Connect.	1-hop	M-hop	Super.	C-Cons.	Avg.
Closed-source LMMs															
GPT-5	93.4	99.4	92.8	95	77.5	64.6	77.5	92.1	92.0	94.4	100	86.7	88.4	88.2	85.6
Gemini-2.5-pro	98.3	100	91.1	95.6	77.5	65.9	74.7	94.4	89.7	82.1	98.3	87.5	87.1	84.3	84.0
Gemini-2.5-flash	97.1	98.3	77.2	91.1	84.2	63.5	66.9	92.6	87.5	69.8	96.7	78.3	90.0	82.9	79.0
Claude-3-7-sonnet-latest	90.4	96.7	59.4	77.8	68.3	60.9	68.4	80.6	81.0	85.2	96.7	70.0	68.6	55.7	74.8
Qwen-VL-Max	76.7	99.4	53.3	83.3	51.7	97.1	53.2	85.7	84.2	33.3	93.3	80.0	65.7	74.3	70.1
GPT-4o	81.3	97.8	46.7	78.3	58.3	94.2	48.2	82.4	76.2	63.9	96.7	80.8	77.1	72.9	70.8
GPT-4.1-mini	78.3	96.7	41.1	68.3	51.7	65.3	57.9	92.6	73.3	53.7	100	75.8	75.7	81.4	67.7
GPT-4o-mini	55.4	93.9	28.9	70.6	35.8	89.7	40.2	72.2	54.3	21.3	96.7	72.5	60.0	64.3	55.9
Qwen-VL-Plus	47.5	71.1	43.3	66.1	11.7	82	44.1	51.9	68.2	19.8	91.7	62.5	38.6	44.3	52.2
Open-source LMMs															
Qwen2.5-VL-72B-instruct	67.1	97.8	51.7	66.1	32.5	60.6	48.7	80.6	80.7	53.4	98.3	68.3	71.4	68.6	63.1
Llama-4-Maverick-17B-128E-Instruct	78.8	91.1	37.8	60.0	31.7	61.1	48.5	75.5	88.1	50.6	98.3	85.8	55.7	67.1	62.8
GLM-4.5V	87.1	96.7	58.3	66.1	53.3	62.4	48.5	87.5	67.2	27.2	95.0	75.8	70.0	74.3	62.7
Llama-4-Scout-17B-16E-Instruct	71.7	87.8	29.4	47.8	37.5	56.9	31.0	73.2	85.5	41.4	98.3	76.7	51.4	52.9	54.5
ERNIE-4.5-VL-28B-A3B-PT	53.2	81.7	43.9	44.4	29.2	59.5	47.8	80.6	52.7	34.3	95.0	67.5	57.1	55.7	53.2
Qwen2.5-VL-7B-instruct	69.2	85.0	49.4	74.4	19.2	56.9	35.3	74.1	52.1	21.0	93.3	59.2	58.6	58.6	51.4
Qwen2.5-omni-7B	65.0	78.3	33.3	44.4	25.0	46.8	30.4	53.2	68.2	18.8	93.3	54.2	60.0	40.0	45.3
Aya-vision-32B	57.5	78.3	34.4	56.7	12.5	47.9	38.0	56.0	35.1	22.5	88.3	58.3	38.6	35.7	43.8
Gemma-3-27B-it	25.4	70.0	28.3	42.8	31.7	51.1	30.7	52.3	65.3	30.9	93.3	57.5	38.6	28.6	42.6
Gemma-3-12B-it	19.2	71.7	13.9	43.9	31.7	54.0	30.0	62.0	42.8	43.8	95.0	63.3	47.1	47.1	42.3

Table: Performance of different LMMs on VKG-QA. G-un., Dire., and Connect. denote graph understanding, relation direction, and connectivity, while 1-hop, M-hop, Super., and C-Cons. correspond to single-hop, multi-hop, superlative, and conditional constraint reasoning.



- Current LMMs still have clear **limitations in understanding and reasoning over visualized KGs**.
- Closed-source models consistently perform better than open-source ones.
- Visual graph representations help models **better capture structural relations** than textualized inputs.
- The main **bottleneck is perceptual grounding of graph structure**, rather than reasoning ability alone.

	Degree	Dire.	Cycle	Connect.	1-hop	M-hop	Super.	C-cons.	Avg.
The results of LLMs									
GPT5-chat	77.4	64.3	98.7	56.6	89.7	91.7	94.0	84.9	77.1
GPT-4o	61.3	91.7	94.7	65.5	87.9	73.3	86.0	84.9	75.1
Qwen3-Max	68.4	90.5	97.3	43.5	98.3	91.7	86.0	84.8	74.6
Qwen-Plus	51.4	86.9	94.7	45.2	91.4	88.3	84.0	84.8	68.7
Qwen2.5-7B-Instruct	24.5	73.8	96.0	55.4	93.1	83.3	78.0	71.7	60.5
Llama3-8B-Instruct	35.9	57.1	96.0	27.4	91.4	66.7	70.0	69.8	53.6
Llama2-7B	14.2	4.8	64.0	22.0	91.4	63.3	60.0	49.1	35.0
The results of LMMs									
GPT-5	82.1	85.7	97.3	94.6	100	85.0	86.0	84.9	88.8
Gemini-2.5-pro	75.9	85.7	98.7	84.5	98.3	88.3	88.0	86.8	85.4
Gemini-2.5-flash	69.3	79.8	96.0	75.0	96.6	85.0	92.0	83.0	80.1
Qwen2.5-VL-72B-instruct	51.9	65.5	96.0	53.6	98.3	85.0	74.0	73.6	67.2
GLM-4.5V	57.6	70.2	97.3	31.6	94.8	83.3	68.0	77.4	64.1

Table: Performance of various LLMs and LMMs on VKG-QA and its variants.